Exercise 2: Vector calculus, conservation laws, classification

Prognostic Modelling (pea700), WiSe 2026/27

Out: week 2 · Back: week 3 · 15 points

Covers unit 2 of the script. Hand in your solutions in the folder `ex02/` of your repository, as PDF or in the markdown cells of a notebook.

Problem 1: Gradient (2 points)

The temperature on a plate is $T(x, y) = 20 - x^2 - 2y^2$ (in °C, x and y in m).

a) Compute the gradient at the point $(1, 1)$ and the unit vector in its direction.

b) In which direction does heat flow at this point according to Fourier's law, and what is the largest rate of change of temperature with distance?

Problem 2: Laplacian (3 points)

Let $f(x, y) = axy + ax^2 + by^3 + y^3$ with constants a and b. Compute $\nabla \cdot (\nabla f)$. For which a and b does f satisfy the Laplace equation everywhere?

Problem 3: Divergence of a radial field (3 points)

Consider the vector field

$$\vec{E} = \frac{\vec{r}}{\|\vec{r}\|^p}, \qquad \vec{r} = x\vec{i} + y\vec{j} + z\vec{k}.$$

a) Derive $\nabla \cdot \vec{E}$ in Cartesian coordinates.

b) For which p is the field free of divergence away from the origin? Which physical fields have this form?

Problem 4: Deriving a conservation law (4 points)

A dissolved substance with concentration $c(x,t)$ (kg/m^3) is transported in a river with constant cross-section A and constant velocity v. It is mixed by turbulence, described as diffusion with coefficient D, and it decays at the rate λc (kg/m^3/s).

a) Write down the mass balance for the river section between x and $x + \Delta x$, and derive the differential equation for c by taking the limit $\Delta x \to 0$.

b) Classify the equation: order, linear or nonlinear, type.

c) Which initial and boundary conditions are needed for a river section of length L? Propose physically meaningful ones.

Problem 5: Classification (3 points)

For each equation state whether it is an ODE or a PDE, its order, whether it is linear, and for the PDEs of second order the type.

1. $\dfrac{\partial u}{\partial t} = D\dfrac{\partial^2 u}{\partial x^2}$
2. $\dfrac{\partial^2 u}{\partial t^2} = c^2\dfrac{\partial^2 u}{\partial x^2}$
3. $\dfrac{\partial^2 \phi}{\partial x^2} + \dfrac{\partial^2 \phi}{\partial y^2} = -\rho(x,y)$
4. $\dfrac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \dfrac{g}{L}\sin\theta = 0$
5. $\dfrac{\partial u}{\partial t} + u\dfrac{\partial u}{\partial x} = \nu\dfrac{\partial^2 u}{\partial x^2}$
6. $\dfrac{\partial \theta(\psi)}{\partial t} = \dfrac{\partial}{\partial z}\left[K(\psi)\left(\dfrac{\partial \psi}{\partial z} + 1\right)\right]$